

MURALI SAI BUDDAKKAGARI VENKATA

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SUMMARY

Applied AI Engineer specializing in LLM agent systems, multi-agent orchestration, and production eval pipelines. Built and shipped agentic workflows using LangGraph, LangChain, and GPT-4o - including a Human-in-the-Loop system completing ~85% of tasks autonomously with full audit trail. Experienced designing evaluation frameworks (RAGAS, faithfulness/relevancy metrics), analyzing failure modes, and iterating on prompts and retrieval strategies to measurably improve agent solve Rate. Researched domain-adaptive ensemble learning for medical nucleus detection (Faster R-CNN, Mask R-CNN, YOLO) at Stevens Institute of technology.

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL, C++, PySpark, TypeScript
- **AI & Machine Learning:** Prompt Engineering & Design, Agent Evaluation (RAGAS), Failure Mode Analysis, Retrieval-Augmented Generation, Multi-Agent Systems, LangChain, LangGraph, LlamaIndex, CrewAI, HuggingFace, Transformers, PyTorch, TensorFlow, Model Context Protocol (MCP), FastAPI, ChromaDB, Scikit-learn, Embedding Models, LLMops, LangSmith, Fine-tuning
- **Infrastructure & DevOps:** AWS (ECS Fargate, S3), Google Cloud Platform, BigQuery, Azure, Docker, PostgreSQL, GitHub Actions, CI/CD, ReactJS, OpenAI API, Claude API

EXPERIENCE

Querkey Inc
Data Science Intern

Jun 2025 - Aug 2025
CA, US

- Engineered end-to-end ML data ingestion pipelines (Python, PySpark, GCP, BigQuery) processing 200+ MP4 files daily, cutting validation latency by 40% and eliminating manual bottlenecks across 3 production data streams.
- Reduced production ML workflow failure rate by 60% by deploying a Dockerized observability stack (MLflow, HuggingFace monitoring, ChromaDB, FastAPI) over 100K+ records, enabling real-time failure detection and proactive incident response for business-critical pipelines.

VIEW Synergy
AI and Automation Intern

Jan 2024 - Apr 2024
KA, IND

- Reduced manual process overhead by ~35% for 100+ employees by architecting AI workflow automation pipelines (LangChain, LlamaIndex, Azure, ReactJS, TypeScript) enabling end-to-end intelligent task orchestration across full SDLC cycles.
- Shipped production RAG pipelines (C#/.NET 8.0, OpenAI API, Docker, SQL) with sub-200ms API latency, supporting reliable agentic tool-use in a regulated Agile delivery environment.

PROJECTS

Agent Orchestration System with Tool Use, Memory, and Human in the Loop

- Engineered a production-grade multi-agent system (LangGraph, GPT-4o, Claude Haiku) with 3-layer agent hierarchy, parallel task execution, and two-tier memory (Redis + ChromaDB), achieving ~85% autonomous task completion rate across long-horizon workflows.
- Designed a Human-in-the-Loop escalation system with 5 automated triggers and 4 approval levels, routing 15% of edge-case tasks to human reviewers with full audit trail - validated across fintech and legal compliance requirements.

SEC EDGAR RAG Filing Analyzer

- Improved RAG eval scores (faithfulness 0.57→0.65, relevancy 0.58→0.68, recall 0.57→0.70) across SEC 10-K filings by iteratively tuning retrieval strategies, chunk sizing, and context construction using LangChain, LlamaIndex, and ChromaDB.
- Deployed investment-banking-grade RAG system with RBAC, MNPI guardrails, and SEC 17a-4/FINRA 4511-aligned audit trail on AWS ECS Fargate (Terraform + GitHub Actions CI/CD) with an automated RAGAS eval pipeline, reducing manual eval effort by ~70% through continuous performance monitoring.

EDUCATION

Stevens Institute of Technology
Master of Science, Applied Artificial Intelligence

May 2026
NJ, US

PES University
Bachelor of Technology, Computer Science (Specialization in ML)

May 2024
KA, IND

PUBLICATIONS & CERTIFICATIONS

- Best Research Paper Award, ICICC 2024 - WorkFit: Real-time Multimodal Posture Monitoring System using CNN-based Computer Vision and Flex Sensor Fusion.